Linux `touch` Command Notes

The `touch` command in Linux is mainly used to **create empty files** or **update the timestamps** of existing files.

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1. Create a New Empty File

```
touch file1.txt
```

Description: Creates an empty file named `file1.txt` if it does not exist.

Example:

```
ls -l
-rw-r--r-- 1 user user 0 Sep 22 10:00 file1.txt
```

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2. Create Multiple Files at Once

```
touch file1.txt file2.txt file3.log
```

Description: Creates multiple empty files in a single command.

Example:

```
ls -l
-rw-r--r-- 1 user user 0 Sep 22 10:00 file1.txt
-rw-r--r-- 1 user user 0 Sep 22 10:00 file2.txt
-rw-r--r-- 1 user user 0 Sep 22 10:00 file3.log
```

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3. Update Timestamp of an Existing File

```
touch file1.txt
```

Description: If `file1.txt` already exists, `touch` updates its **last access** and **modification time**.

Example:

```
ls -l
-rw-r--r-- 1 user user 0 Sep 22 11:30 file1.txt
```

(time updated without changing file content)

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4. Do Not Create New File (`-c`)

```
touch -c fileX.txt
```

Description: Updates timestamp only if file exists. Does not create a new file.

Example: - If `fileX.txt` exists → timestamp updated. - If `fileX.txt` does not exist → no file created.

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5. Set a Custom Date and Time (`-t`)

```
touch -t 202509220930 file1.txt
```

Description: Sets file timestamp to **22 Sep 2025, 09:30**.

Format: `[[CC]YY]MMDDhhmm[.ss]`

Example:

```
ls -l
-rw-r--r-- 1 user user 0 Sep 22 09:30 file1.txt
```

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6. Use Reference File for Timestamp (`-r`)

```
touch -r ref.txt target.txt
```

Description: Sets `target.txt` timestamp same as `ref.txt`.

Example:

```
ls -l ref.txt target.txt
-rw-r--r-- 1 user user 0 Sep 21 15:00 ref.txt
-rw-r--r-- 1 user user 0 Sep 21 15:00 target.txt
```

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7. Create Hidden Files

```
touch .hiddenfile
```

Description: Creates a hidden file (starts with `.`).

Example:

```
ls -a
.  ..  .hiddenfile  file1.txt
```

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8. Combine with `ls -la`

```
touch report.txt notes.md .hidden
ls -la
```

Output:

```
-rw-r--r-- 1 user user 0 Sep 22 10:10 report.txt
-rw-r--r-- 1 user user 0 Sep 22 10:10 notes.md
-rw-r--r-- 1 user user 0 Sep 22 10:10 .hidden
```

Quick Summary

- `touch file.txt` → Create new empty file
- `touch f1 f2 f3` → Create multiple files
- `touch file.txt` → Update timestamp of existing file
- `touch -c file.txt` → Update timestamp only if file exists
- `touch -t YYYYMMDDhhmm file.txt` → Set custom time
- `touch -r ref.txt target.txt` → Copy timestamp from another file
- `touch .hiddenfile` → Create hidden file

✅ `touch` is mostly used for **quick file creation** and **modifying timestamps**.